Evaluation of LQR and MPC Controllers for Networked Control Systems

Ricardo A. Alayon M.
Master of Science in Information and Communication Sciences
Universidad Distrital Francisco José de Caldas
Bogotá, Colombia
raalayonm@udistrital.edu.co

Andres L. Jutinico
Electronic Engineering Program
Master of Science in Information and Communication Sciences
Universidad Distrital Francisco José de Caldas
Bogotá, Colombia
aljutinicoa@udistrital.edu.co

Oscar D. Montoya
Electrical Engineering Program
Universidad Distrital Francisco José de Caldas
Bogotá, Colombia
odmontoyag@udistrital.edu.co

***Abstract*—Wireless Networked Control Systems (NCS) are affected by communication issues such as packet losses, which can degrade closed-loop performance and stability. This paper presents a comparison between the Linear Quadratic Regulator (LQR) and Model Predictive Control (MPC) for packet loss compensation in a control system with a finite-state Markov communication channel. Both controllers are evaluated under the same dynamic system and channel conditions through simulations, using root mean square (RMS) values to compare tracking performance and control effort. The MPC strategy uses predictive control actions by transmitting the current input and future predictions in each packet, enabling the compensation of short packet-loss bursts. Simulations of the control system show that LQR maintains stability but experiences higher degradation in velocity tracking as channel reliability decreases. In contrast, MPC preserves closer-to-nominal performance. Finally, experimental tests with a Quanser QUBE motor confirm the capability of MPC for packet loss compensation under sensor noise and degraded communication conditions.**

***Index Terms*—Linear Quadratic Regulator (LQR), Model Predictive Control (MPC), control over wireless networks, finite-state Markov chain.**

I. Introduction

Networked control systems (NCS) have become increasingly relevant as wireless communications become widespread in modern engineering applications. Systems such as autonomous vehicles, mobile robots, sensor networks, and unmanned aerial vehicles rely on wireless links to exchange and process information between sensors, controllers, actuators, and even human operators. Although replacing wired communication with wireless technologies enables greater mobility and scalability, it also introduces significant challenges related to reliability and performance. In particular, wireless communication channels may be affected by stochastic phenomena such as packet losses and time-varying connectivity. These effects can degrade closed-loop performance, affect stability, and reduce control accuracy in feedback systems. Consequently, the design of control strategies must explicitly account for communication uncertainties.

A common approach to modeling these uncertainties is through Markov communication channels, which describe the random evolution of packet transmission success or failure over time. In this context, the interaction between the controller and the plant is influenced by the probabilistic nature of the communication medium, leading to hybrid system behavior.

Several control strategies are commonly used in the literature, including the Linear Quadratic Regulator (LQR) and Model Predictive Control (MPC). Although these methods are well-established in deterministic scenarios, their performance under stochastic communication constraints remains an active research topic. This paper presents a comparative analysis of LQR and MPC applied to a closed-loop system with a Markovian communication channel. The study evaluates the impact of channel-induced packet losses on closed-loop performance and compares the performance of both control strategies under these conditions. Transmission delays and external disturbances are intentionally excluded to isolate the impact of the Markovian packet-loss mechanism.

Networked control systems (NCS) have been extensively studied under communication constraints such as delays, packet dropouts, and channel uncertainties. Discrete-time Markov jump linear systems provided a general framework for modeling systems with random mode changes [1]. Such models can also be useful for describing switching behaviors in networked control systems, while robust control strategies have addressed state delays and parametric uncertainties to mitigate performance degradation [2], [3]. More recently, communication-aware approaches have used Markovian wireless channel models to represent time-varying reliability and packet loss, enabling optimal and robust control designs with packet loss compensation [4], [5]. In addition, Lyapunov-based methods have provided mean-square stability guarantees for Markovian switching systems under network-induced uncertainties [6], [7].

In terms of estimation and filtering, recursive and Kalman-based approaches have been proposed to accommodate Marko-

vian jump dynamics and random delays [8], [9]. These methods have focused on uncertainty propagation through the communication network and its effect on state reconstruction accuracy. Additionally, secure and reliable state estimation over Markov channels has been investigated in [10], where communication constraints are explicitly incorporated into the estimation process.

Control over systems conditioned by communication has also been addressed from an optimization perspective. In particular, linear quadratic regulation under network restrictions has attracted researchers' attention, with results highlighting relevant performance limitations in learning-based and optimal control frameworks [11]. Extensions to more complex dynamical structures, including quadrotor systems under external disturbances, further illustrate the relevance of robust and intelligent control design in uncertain environments [12].

Despite these advances, the interaction between classical optimal control techniques and communication channels modeled by Markov processes remains an active area of research. Namely, the combined effect of controller structure and channel-induced stochasticity continues to pose challenges in analyzing and predicting performance degradation caused by communication uncertainties, while ensuring system performance, stability, and robustness. In this work, LQR and MPC are comparatively evaluated under packet loss conditions modeled by a finite-state Markov channel. The analysis considers RMS measurements to quantify tracking performance and control effort. Furthermore, MPC is experimentally evaluated as a packet loss compensation strategy under sensor noise, considering its predictive capability to transmit future control actions and improve robustness against communication losses.

II. Methodology

A. Model Formulation

Consider a discrete-time closed-loop control scheme, as shown in Fig. 1, where the feedback loop is implemented over a wireless communication channel modeled as a Markov process. In this context, the feedback signal may not be transmitted reliably due to packet losses, which could directly affect the performance and stability of the closed loop. To analyze this effect, the discrete-time model is considered:

$$\mathbf{x}_{k+1} = \mathbf{F}\mathbf{x}_k + \mathbf{G}u_k, \tag{1}$$

where $\mathbf{x}_k \in \mathbb{R}^n$ represents the state vector, u_k is the control input, and $\mathbf{F}, \mathbf{G}$ are the discrete-time representations of the system matrices $\mathbf{A}$ and $\mathbf{B}$ shown in Fig. 1.

When the feedback loop is closed through the wireless channel, the control input is affected by a binary stochastic process $S_k \in \{0, 1\}$, where $S_k = 1$ represents the bad state (susceptible to packet loss) and $S_k = 0$ represents the good state (where successful transmission during each sample time T_s is highly probable). Under this consideration, we define the following hybrid model combining the dynamic system and the communication channel:

$$\mathbf{x}_{k+1} = \mathbf{F}\mathbf{x}_k + \mathbf{G}(1 - S_k)u_k. \tag{2}$$

The evolution of S_k is modeled as a Markov process, commonly employed to represent packet loss bursts as observed in wireless communication channels due to the persistence of adverse conditions [10]. Fig. 1 illustrates the resulting networked control architecture, where the controller, process plant, and communication channel form a hybrid system.

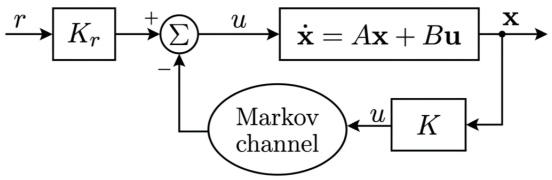


Fig. 1. Block diagram for the hybrid system.

B. Markovian Communication Channel

The feedback signal is transmitted through a channel modeled using a two-state Gilbert–Elliott process. The binary variable S_k denotes the channel state, where $S_k = 0$ represents the good state and $S_k = 1$ represents the bad state. The channel state evolves according to a Markov process, introducing temporal correlation and bursty packet losses. The equivalent channel is modeled as

$$\mathbf{y}_k = \mathbf{x}_k(1 - e_k), \tag{3}$$

where e_k is the realization of the Markov state process S_k ($e_k = 0$ for good, $e_k = 1$ for bad). The state transitions follow

$$P(S_{k+1} = j \mid S_k = i) = p_{ij}, \tag{4}$$

with the transition matrix defined as

$$\mathbf{P} = \begin{bmatrix} p_{GG} & p_{GB} \\ p_{BG} & p_{BB} \end{bmatrix}. \tag{5}$$

Since $p_{GG} = 1 - p_{GB}$ and $p_{BB} = 1 - p_{BG}$, the transition probabilities are complementary. In this work, simulations are parameterized by p_{GG} (the probability of remaining in the good state), whereas experiments are characterized by p_{GB} (the transition probability from the good to the bad state). The experimental values are set to:

$$p_{GB} \in \{0.01, 0.02, 0.03, 0.04, 0.05, 0.1, 0.15, 0.2, 0.25, 0.3\}.$$

This range, adopted from [13], covers communication conditions from nearly ideal to substantial packet loss.

C. Control Strategies

The foundations of optimal control theory are closely linked to the works of Bellman on dynamic programming [14] and Pontryagin's maximum principle [15], which established the mathematical foundation of modern optimal control. These ideas were later formalized in the context of linear-quadratic regulation, particularly through Kalman's contributions [16] and the comprehensive treatment by Anderson and Moore [17]. Together, these works define the classical LQR framework, which remains widely used in state-feedback design.

Model Predictive Control (MPC) originated from early industrial and heuristic approaches to constrained control [18], [19]. Its later theoretical development provided rigorous guarantees of stability and optimality, as established in the works of Rawlings and Mayne [20] and Mayne et al. [21]. This progression made MPC a standard methodology for constrained multivariable systems, particularly in applications where prediction and constraint handling are essential.

Readers familiar with control theory will recognize these results as standard, while others are referred to the cited works for a more complete and rigorous treatment. The controller designs adopted in this study follow these well-established formulations and are used without further modification due to their maturity in the literature. Based on this framework, the following sections present the simulation setup and analyze the results obtained under the considered communication constraints.

D. Model and Parameters for Control Design

A DC motor was used as the plant. The considered states are the armature current (i) and angular velocity (ω), while the control input is the motor voltage u. The continuous-time dynamics are given by:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t), \quad \mathbf{A} = \begin{bmatrix} -\frac{R}{L} & -\frac{k_m}{L} \\ \frac{k_m}{J} & -\frac{b}{J} \end{bmatrix}, \mathbf{B} = \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix}. \tag{6}$$

The model parameters were based on the Quanser QUBE DC motor [22]. Table I lists the complete set of physical, computational, and required parameters necessary to successfully perform the simulations.

LQR and MPC were selected because both are model-based control strategies. In the unconstrained case with an infinite prediction horizon ($N \to \infty$), MPC is closely related to the discrete-time LQR formulation [23]. Table I summarizes the controller parameters. MPC uses a control horizon of three samples and a prediction horizon of ten, whereas LQR uses the weighting matrices specified in the table. Their main difference under packet losses is that MPC transmits the current control action and future predictions, allowing previously received actions to be executed during up to two consecutive sampling-period losses using a small buffer. In contrast, LQR relies solely on the current control action. The MPC input was left unconstrained because the computed actions remained within the physically admissible range.

E. Simulation Performance Evaluation

The performance of each control strategy is evaluated under identical initial conditions and reference signal. Each simulation generates current, velocity, and control input to characterize the system response. Also, the state of the Markovian channel, where 1 is packet losses. The analysis highlights both the performance degradation introduced by the communication channel and the distinct responses of each controller. Dashed red traces represent the ideal loss-free performance and are used as a reference for comparison with closed-loop simulations conducted over the wireless channel. RMS values of

TABLE I
COMPLETE SIMULATION AND CONTROLLER PARAMETERS

Category	Parameter / Description	Value
Motor (QUBE)	Armature resistance (R)	$6.3\,\Omega$
	Armature inductance (L)	$0.00085\,\mathrm{H}$
	Back-emf constant (k_m)	$0.036\,\mathrm{V \cdot s/rad}$
	Rotor inertia (J)	$4.0 \times 10^{-6}\,\mathrm{kg \cdot m^2}$
	Viscous damping (b)	$5.7 \times 10^{-4}\,\mathrm{N \cdot m \cdot s/rad}$
Open-Loop Test	Step input voltage (u)	$4.0\,\mathrm{V}$
Simulation Setup	Sampling period (T_s)	$0.001\,\mathrm{s}$
	Simulation time (t_{fin})	$2\,\mathrm{s}$
	Samples (N)	2000
	Reference speed (r)	$3.0\,\mathrm{rad/s}$
	Initial state ($\mathbf{x}_0$)	$[0 \quad 0]^T$
MPC	Control horizon (m)	3
	Prediction horizon (p)	10
	Actuator constraints (MV)	$[-\infty, \infty]\,\mathrm{V}$
LQR	Weighting matrix (Q)	$\mathrm{diag}(50000, 7)$
	Weighting factor (R)	1

current, velocity, and control input are calculated for different packet loss probabilities, providing a quantitative measure of system performance.

Figs. 2 and 3 illustrate the effects of the communication channel by using identical error sequences for the same probability values, enabling a direct comparison of transient behavior. Simulations and RMS calculations were performed over 2 seconds. Closing the loop through the wireless link introduces fluctuations caused by packet losses. With the LQR controller, the system remains stable; however, velocity tracking is significantly affected by information losses. Under persistent errors, the trajectory deviates from the reference, as shown in Fig. 2, where p_{GG} denotes the probability of remaining in the good channel state, and $p_{GG} = 70\%$ represents the most adverse simulated channel condition considered.

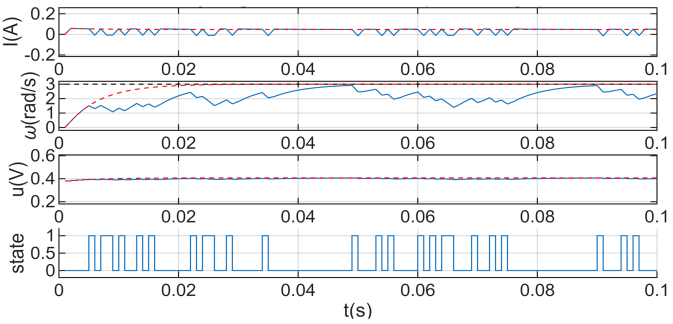


Fig. 2. System evolution for LQR controller with probability ($p_{GG} = 70\%$).

In the Fig. 3 simulation, MPC exhibits a rapid settling response with limited oscillations. At first glance, predictive control reduces the noisy behavior seen with LQR, demonstrating its robustness under channel errors.

III. ANALYSIS OF SIMULATIONS OF THE CONTROL SYSTEM

The simulated system exhibits degraded performance when the control loop is closed through the communication channel

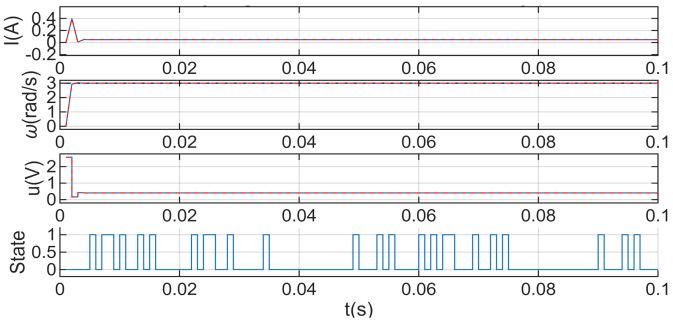


Fig. 3. System evolution for MPC controller with probability ($p_{GG} = 70\%$).

compared with the "wired" system due to transmission errors. The trajectories vary according to the different error probabilities. For LQR, clear differences are observed in current and velocity responses. The controller is sensitive to channel state transitions, producing deviations from the reference values that increase as errors persist. MPC shows clear compensation improvement, with control signals rapidly stabilizing and minimal oscillations. The trajectories remain close to the reference during the observed time window, demonstrating its robustness under adverse channel conditions. Based on this analysis, LQR presents the poorest performance, whereas MPC achieves more suitable reference tracking. The RMS analysis provides quantitative support for this evaluation.

A. RMS Analysis

This section analyzes the numerical RMS measurements to confirm the conclusions obtained from the previous section plots, including the *"worst cases"* presented in Section II.

The RMS values of current, velocity, and control input were calculated for each channel condition. For each variable, the corresponding Δ value represents the difference between its RMS under the considered channel condition and its RMS in the nominal, loss-free case ($p_{GG} = 100\%$). Thus, Δ quantifies the change in signal RMS relative to the nominal response. These values are used to compare how each controller responds to decreasing channel reliability; they do not directly represent the tracking error with respect to the reference.

The LQR results show a progressive reduction in velocity RMS as p_{GG} decreases, since uncompensated dropouts cause open-loop drift and energy loss in the speed response. From $p_{GG} = 100\%$ to 70%, the current and control RMS values also decrease slightly. These changes indicate that packet losses affect the velocity response more noticeably than the RMS magnitudes of the current and control signals. Since RMS measures signal magnitude rather than deviation from the reference, these values are interpreted as indicators of changes in system response, not as direct tracking-error measurements. Table II shows the RMS measurements from simulations performed whith LQR.

The control signal u shows minor sensitivity to packet losses, reflecting subtle differences in the control effort required to compensate for channel uncertainties. Table III shows the RMS measurements from simulations performed with the MPC controller.

TABLE II
LINEAR QUADRATIC REGULATOR (LQR) PERFORMANCE MEASURES UNDER MARKOVIAN PACKET DROPOUT

Chan.	**Current I (A)**		**Speed ω (rad/s)**		**Control u (V)**	
p_{GG}	**RMS**	Δ	**RMS**	Δ	**RMS**	Δ
100%	0.047523	–	2.993309	–	0.407171	–
99%	0.047424	-0.000099	2.964727	-0.028582	0.406903	-0.000268
98%	0.047276	-0.000247	2.924353	-0.068956	0.406524	-0.000647
97%	0.047129	-0.000394	2.881188	-0.112121	0.406129	-0.001042
96%	0.046879	-0.000644	2.818399	-0.174910	0.405541	-0.001630
95%	0.046892	-0.000631	2.823514	-0.169795	0.405584	-0.001587
90%	0.046118	-0.001405	2.648143	-0.345166	0.403948	-0.003223
85%	0.045722	-0.001801	2.553493	-0.439816	0.403105	-0.004066
80%	0.044763	-0.002760	2.369091	-0.624218	0.401423	-0.005748
75%	0.044328	-0.003195	2.298874	-0.694435	0.400759	-0.006412
70%	0.043946	-0.003577	2.224872	-0.768437	0.400125	-0.007046

TABLE III
MODEL PREDICTIVE CONTROL (MPC) PERFORMANCE MEASURES UNDER MARKOVIAN PACKET DROPOUT

Chan.	**Current I (A)**		**Speed ω (rad/s)**		**Control u (V)**	
p_{GG}	**RMS**	Δ	**RMS**	Δ	**RMS**	Δ
100%	0.049027	–	2.994560	–	0.414337	–
99%	0.049027	0.000000	2.994560	0.000000	0.414337	0.000000
98%	0.050120	0.001093	2.987444	-0.007116	0.419309	0.004972
97%	0.051121	0.002094	2.981026	-0.013534	0.423826	0.009489
96%	0.052908	0.003881	2.968349	-0.026211	0.431940	0.017603
95%	0.052709	0.003682	2.969731	-0.024829	0.430929	0.016592
90%	0.057067	0.008040	2.937240	-0.057320	0.451300	0.036963
85%	0.058951	0.009924	2.923461	-0.071099	0.460332	0.045995
80%	0.062939	0.013912	2.890254	-0.104306	0.479684	0.065347
75%	0.064008	0.014981	2.881469	-0.113091	0.484986	0.070649
70%	0.065326	0.016299	2.871238	-0.123322	0.491590	0.077253

For MPC, the velocity RMS remains closer to its nominal value as the probability of remaining in the good channel state decreases. From $p_{GG} = 100\%$ to 70%, it decreases from 2.995 rad/s to 2.871 rad/s while the current and control RMS values increase. This behavior reflects active loss compensation via control predictions, trading an 18.6% increase in control effort to limit velocity RMS degradation to just 4.1% (compared to 25.7% under LQR). Although these RMS measurements do not directly quantify tracking error, the smaller change in velocity RMS compared with LQR indicates that the velocity signal is less affected by the tested channel conditions under MPC.

Although speed control is the main objective, current variations are analyzed to provide a global view of the system response and the coupling between the control input and system dynamics. Overall, MPC demonstrates higher robustness, maintaining nearly constant velocity RMS performance under severe channel degradation at the cost of increased control effort.

IV. EXPERIMENTAL RESULTS USING MPC

Experimental tests were conducted on the Quanser QUBE motor [22] using the same parameters as the MPC simula-

tions. For each channel condition, the packet-loss sequence generated by the Markov model was stored and reused in the corresponding simulation and experiment. The same variables were measured, and RMS values were computed to compare the responses quantitatively. The experimentally measured RMS values are presented in the Table IV.

For comparison, the nominal case was also evaluated. Fig. 4 shows the nominal response, while Fig. 5 provides a detailed view over a 3-second interval.

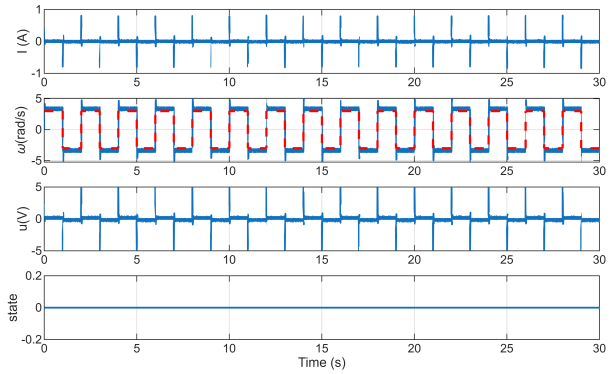


Fig. 4. Wired case during 30 seconds.

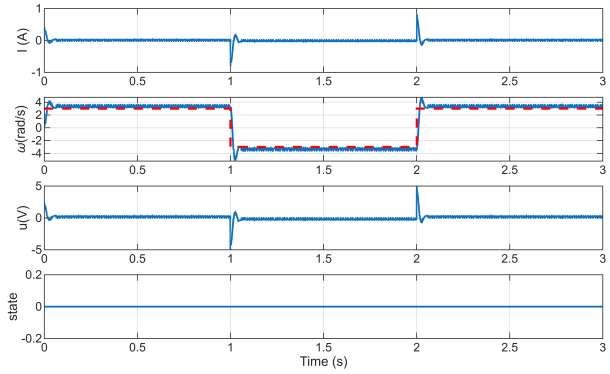


Fig. 5. Detail of wired case ($p_{GG} = 100\%$), 3 seconds.

For the wired case, the measured response fluctuates around the reference due to motor sensor noise, revealing differences between the simulation model and the experimental setup.

TABLE IV
RMS VALUES FOR DIFFERENT OPERATING CONDITIONS

Variable	Nominal	95%	70%
Current i (A)	0.066391	0.066448	0.065111
Speed ω (rad/s)	3.326874	3.166074	2.719227
Control u (V)	0.447578	0.445449	0.432157

Under near-ideal channel conditions, the same noisy behavior is observed, and motor velocity deviate from the reference, as shown in Fig. 6. Figure 7 shows the first 3 seconds for detailed analysis. For a good-to-bad transition probability of $p_{GB} = 5\%$, noise effects become more significant. The speed response is particularly affected by error bursts; when the channel remains in a bad state, the speed deviates from the reference. Direction changes reveal the higher compensation effort required by MPC and the longer recovery time needed to restore the reference.

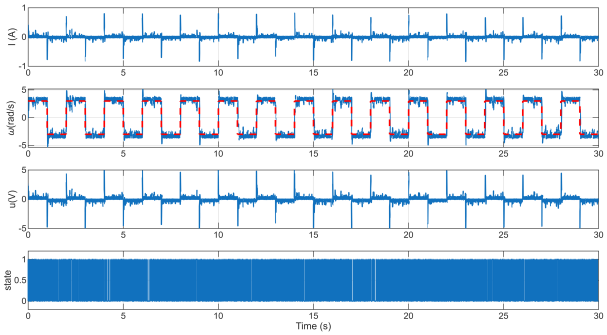


Fig. 6. Experimental response for $p_{GG} = 95\%$.

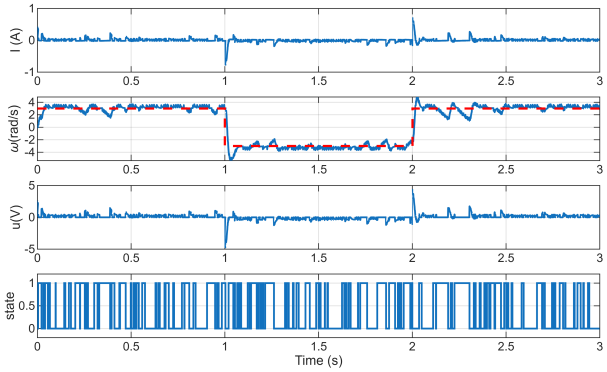


Fig. 7. Experimental response for $p_{GG} = 95\%$ (first 3 seconds).

The most adverse tested condition, shown in Figs. 8 and 9, illustrates MPC behavior under a good-to-bad transition probability of $p_{GB} = 30\%$. Error bursts significantly affect the speed response, increasing noise around the reference compared to $p_{GB} = 5\%$.

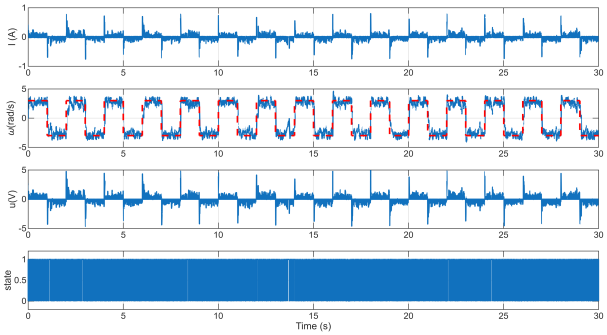


Fig. 8. Experimental response for $p_{GG} = 70\%$.

V. DISCUSSION

The simulation results show that MPC exhibits a smaller variation in velocity RMS than LQR under the tested Marko-

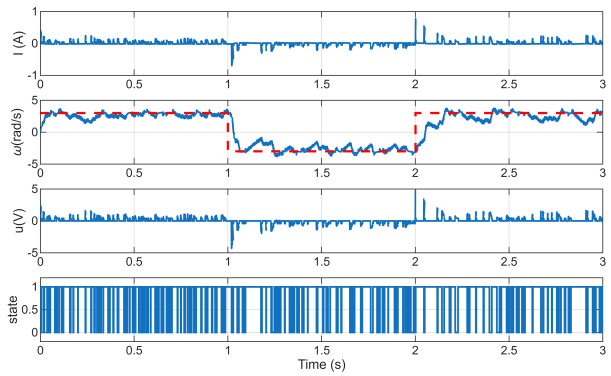


Fig. 9. Experimental response for $p_{GG} = 70\%$ (first 3 seconds).

vian channel conditions, at the cost of increased control effort. As p_{GG} decreases from 100% to 70%, velocity RMS decreases from 2.993 to 2.225 rad/s for LQR and from 2.995 to 2.871 rad/s for MPC. These results indicate that MPC maintains velocity RMS closer to its nominal value as the probability of remaining in the good channel state decreases.

The experimental evaluation extends the analysis to the physical plant using MPC. Fluctuations appear even in nominal conditions due to noise and physical differences, yielding a speed RMS of 3.3269 rad/s compared to 2.9946 rad/s in simulation. Under packet loss, extra deviations occur; at 70% availability, the measured speed (2.7192 rad/s) follows the simulated trend (2.8712 rad/s). These results complement the simulation study on the physical plant.

VI. Conclusion

This paper compared LQR and MPC for packet-loss compensation in a networked control system modeled by a finite-state Markov channel. Under the tested simulation conditions, MPC exhibited a smaller variation in velocity RMS than LQR as p_{GG} decreased from 100% to 70%, at the cost of increased current and control RMS values. This behavior is consistent with the use of previously transmitted future control actions during packet losses. Experimental tests on the Quanser QUBE motor further examined MPC using replayed packet-loss sequences. Despite measurement noise and deviations from the nominal response, the results illustrate the controller's behavior under the tested conditions.

In future work, we will attempt to incorporate the robust Markovian Kalman filter used in [24], [25], to predict the hybrid system states and transmit packets, including potentially lost measurements, to the control vector, thereby addressing uncertainty and noise.

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